

Equity Dilution Protection and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Protection (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.53 (0.42), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 3.07 (2.47), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 19 bps/month with a t-statistic of 2.75.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantly reflect all available information as predicted by traditional efficient market theory. While the efficient markets hypothesis remains a cornerstone of financial economics, researchers increasingly recognize that informational frictions and investor inattention create predictable patterns in asset returns. This paper examines a novel signal—Equity Dilution Protection (EDP)—that captures firms’ ability to shield existing shareholders from the dilutive effects of new equity issuance through mechanisms such as share buybacks and restricted issuance policies. Despite the economic importance of dilution for shareholder value, market participants appear to incorporate information about firms’ dilution protection strategies only gradually. We document that EDP strongly predicts cross-sectional stock returns, with firms exhibiting stronger dilution protection subsequently outperforming those with weaker protection. This predictability concentrates in market segments where information processing constraints bind most tightly, suggesting that slow information diffusion drives the return patterns we observe.

The gradual incorporation of dilution protection information into stock prices reflects fundamental limits on investor attention and information processing capacity. [Hong et al. \(2000\)](#) demonstrate that firm-specific news diffuses slowly across the investor population, particularly for stocks with low analyst coverage, creating momentum in returns as information gradually reaches marginal investors. Similarly, [DellaVigna and Pollet \(2009\)](#) show that investors exhibit limited attention to information released when competing stimuli are high, leading to delayed price responses. In the context of equity dilution protection, these informational frictions become particularly pronounced because evaluating a firm’s true dilution risk requires processing complex information scattered across financial statements, corporate announcements, and governance documents. [Hirshleifer et al. \(2009\)](#) argue that investors face cog-

nitive constraints that prevent them from fully processing all available information simultaneously, leading to systematic underreaction to value-relevant signals. The complexity of assessing dilution protection strategies—which involve understanding share repurchase programs, employee compensation structures, and capital raising alternatives—exacerbates these attention constraints. When firms implement effective dilution protection measures, the positive implications for existing shareholders emerge gradually as sophisticated investors first recognize the value implications, followed by broader market participants. [Cohen and Frazzini \(2008\)](#) provide evidence that even seemingly straightforward accounting information experiences delayed incorporation into prices due to investor inattention, suggesting that more complex signals like EDP should exhibit even stronger patterns of gradual diffusion. We therefore predict that firms with strong equity dilution protection will experience positive return continuation as information about their shareholder-friendly policies slowly permeates the market. This effect should be most pronounced among smaller firms with limited analyst coverage, where [Hong and Stein \(1999\)](#) show that information diffusion occurs most slowly, and among stocks with lower institutional ownership, where professional information processing resources are scarce.

Our empirical analysis reveals that equity dilution protection strongly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on EDP achieving an annualized gross Sharpe ratio of 0.53 and a monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 22 basis points per month with a t-statistic of 3.07. The strategy’s performance remains robust after accounting for transaction costs, generating an annualized net Sharpe ratio of 0.42 and a monthly net alpha of 17 basis points relative to the six-factor model with a t-statistic of 2.47. Notably, the EDP signal maintains significant predictive power even after controlling for the six most closely related strategies from the factor zoo, producing a gross monthly alpha of 19 basis points with a t-

statistic of 2.75 when simultaneously controlling for Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, and change in net operating assets. The return predictability exhibits striking variation across size quintiles, consistent with our hypothesis that information diffusion occurs more slowly for stocks with greater informational frictions. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EDP strategy achieves an average return of 33 basis points per month with a t-statistic of 3.81, while alphas relative to the five most common factor models range from 26 to 39 basis points per month with t-statistics between 2.96 and 4.38. The economic magnitude of these results underscores the importance of dilution protection information for asset pricing. A dollar invested in the EDP strategy would have grown to \$3.40 on a gross basis and \$2.09 after accounting for transaction costs, placing the strategy in the top 6% of all documented anomalies in terms of cumulative performance over our sample period from 1973 to 2024.

This paper makes several contributions to the literature on cross-sectional return predictability and market efficiency. First, we extend the work of [Pontiff and Woodgate \(2008\)](#) on share issuance and returns by identifying a novel dimension of equity market activity—dilution protection—that has not been previously examined in the asset pricing literature. While [Loughran and Ritter \(1995\)](#) document negative returns following equity offerings and [Ikenberry et al. \(1995\)](#) find positive returns following share repurchases, our EDP measure provides a comprehensive signal that captures firms’ net dilution protection stance and predicts returns beyond these individual corporate events. Second, we contribute to the growing literature on slow information diffusion by providing new evidence that complex, multifaceted corporate policies experience particularly gradual price incorporation. Our findings complement [Cohen and Lou \(2012\)](#) who show that supplier-customer links predict returns due to gradual information transmission, but we focus on within-firm infor-

mation that should theoretically be immediately available to all market participants. The concentration of EDP’s predictive power among larger, more liquid stocks challenges conventional wisdom about market efficiency and suggests that informational frictions extend beyond the small-cap universe typically associated with limits to arbitrage. Third, our results have important implications for corporate finance theory and practice. The strong return predictability associated with equity dilution protection suggests that managers’ decisions regarding share issuance and repurchase policies have first-order effects on firm value that markets only gradually recognize. This finding relates to [Baker et al. \(2003\)](#) who argue that market timing motivations drive corporate financing decisions, but we show that the market’s slow recognition of dilution protection strategies creates additional incentives for managers to signal their commitment to protecting existing shareholders. The persistence of the EDP anomaly over our five-decade sample period, combined with its robustness to transaction costs and risk adjustments, suggests that informational frictions create lasting impediments to market efficiency that sophisticated investors can exploit.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our analysis utilizes annual fundamental data items to construct the Equity Dilution Protection signal. The sample includes all firms with available data for the required variables, spanning several decades of financial reporting. construction of our signal relies on two key COMPUSTAT variables. Long-term debt issuance (DLTIS) represents the dollar amount of long-term debt issued by the firm during the fiscal year, capturing new borrowing activities with maturities exceeding one year. This item reflects the cash proceeds from issuing bonds, notes, and other long-term debt

instruments. Stockholders' equity (SEQ) measures the total common and preferred stockholders' equity reported on the balance sheet, representing the residual interest in the firm's assets after deducting liabilities. We construct the Equity Dilution Protection signal by computing the year-over-year change in long-term debt issuance scaled by lagged stockholders' equity: $(DLTIS(t) - DLTIS(t-1)) / SEQ(t-1)$. This measure captures the change in a firm's debt financing activity relative to its equity base. An increase in this ratio indicates that the firm is accelerating its long-term debt issuance relative to its equity capital, potentially signaling management's preference for debt financing over equity issuance to fund operations or investments. We calculate this signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both DLTIS and SEQ, and exclude observations where lagged stockholders' equity is zero or negative to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-sectional mean (median) EDP is -0.29 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal's interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.27% per month with a t-statistic of 3.81. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 3.07 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.30, with a t-statistic of 6.35 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 551 stocks and an average market capitalization of at least \$1,439 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 23 bps/month with a t-statistics of 4.86. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -3-24bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.48. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and al-

phas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDP strategy achieves an average return of 33 bps/month with a t-statistic of 3.81. Among these large cap stocks, the alphas for the EDP strategy relative to the five most common factor models range from 26 to 39 bps/month with t-statistics between 2.96 and 4.38.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy’s gross (net) Sharpe ratio of 0.53 (0.42) is greater than 92% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDP strategy (red line).² Ignoring trading costs, a \$1 invested in the EDP strategy would have yielded \$3.40 which ranks the EDP strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDP strategy would have yielded \$2.09 which ranks the EDP strategy in the top 6% across the 212 anomalies.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 64 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 78 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed 1.16, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on EDP is 0.03.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies

individually, the EDP strategy earns alphas that range from 18-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.67, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.75.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes EDP grows to \$1183.39.

8 Conclusion

This study provides robust evidence that Equity Dilution Protection (EDP) represents a significant and economically meaningful signal for predicting cross-sectional

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDP is available.

stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDP-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on EDP delivers an annualized net Sharpe ratio of 0.42, which remains economically significant after accounting for implementation costs. More importantly, the strategy generates statistically significant alpha relative to the Fama-French five-factor model plus momentum, with net returns of 17 basis points per month (t -statistic = 2.47). The signal’s robustness is further validated by its ability to maintain a significant alpha of 19 basis points per month (t -statistic = 2.75) even after controlling for six closely related financing and growth-based factors, including net debt financing, asset growth, and changes in net operating assets. This orthogonality to existing factors suggests that EDP captures a distinct dimension of the cross-section of expected returns not fully explained by traditional risk factors or related corporate financing activities.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The persistence of the EDP premium after transaction costs indicates that the signal may be exploitable in real-world portfolio construction, particularly for institutional investors with access to efficient execution platforms. The economic magnitude of the returns, combined with the moderate Sharpe ratio, suggests that EDP could serve as a valuable addition to multi-factor investment strategies, potentially improving portfolio risk-adjusted returns through its low correlation with existing factors.

However, this study is subject to several limitations that warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe of stocks examined may influence the gen-

eralizability of our findings. The effectiveness of EDP as a predictive signal may vary across different market regimes, particularly during periods of significant market stress or changing monetary policy environments. Additionally, as with any empirical asset pricing study, there remains the possibility that the documented premium represents compensation for an unidentified systematic risk factor rather than a market inefficiency.

Future research should explore several promising avenues to extend our understanding of the EDP signal. First, investigating the economic mechanisms underlying the predictive power of EDP would provide valuable insights into why markets appear to misprice firms based on their equity dilution characteristics. This could involve examining the role of investor attention, limits to arbitrage, or behavioral biases in driving the EDP premium. Second, extending the analysis to international markets would test the external validity of our findings and determine whether the EDP effect is a global phenomenon or specific to certain market structures. Third, examining the time-varying nature of the EDP premium and its relationship with macroeconomic conditions could reveal important insights about when the strategy is most effective. Finally, investigating potential interactions between EDP and corporate governance characteristics or management quality metrics might uncover additional sources of return predictability and help refine the signal's implementation in practice.

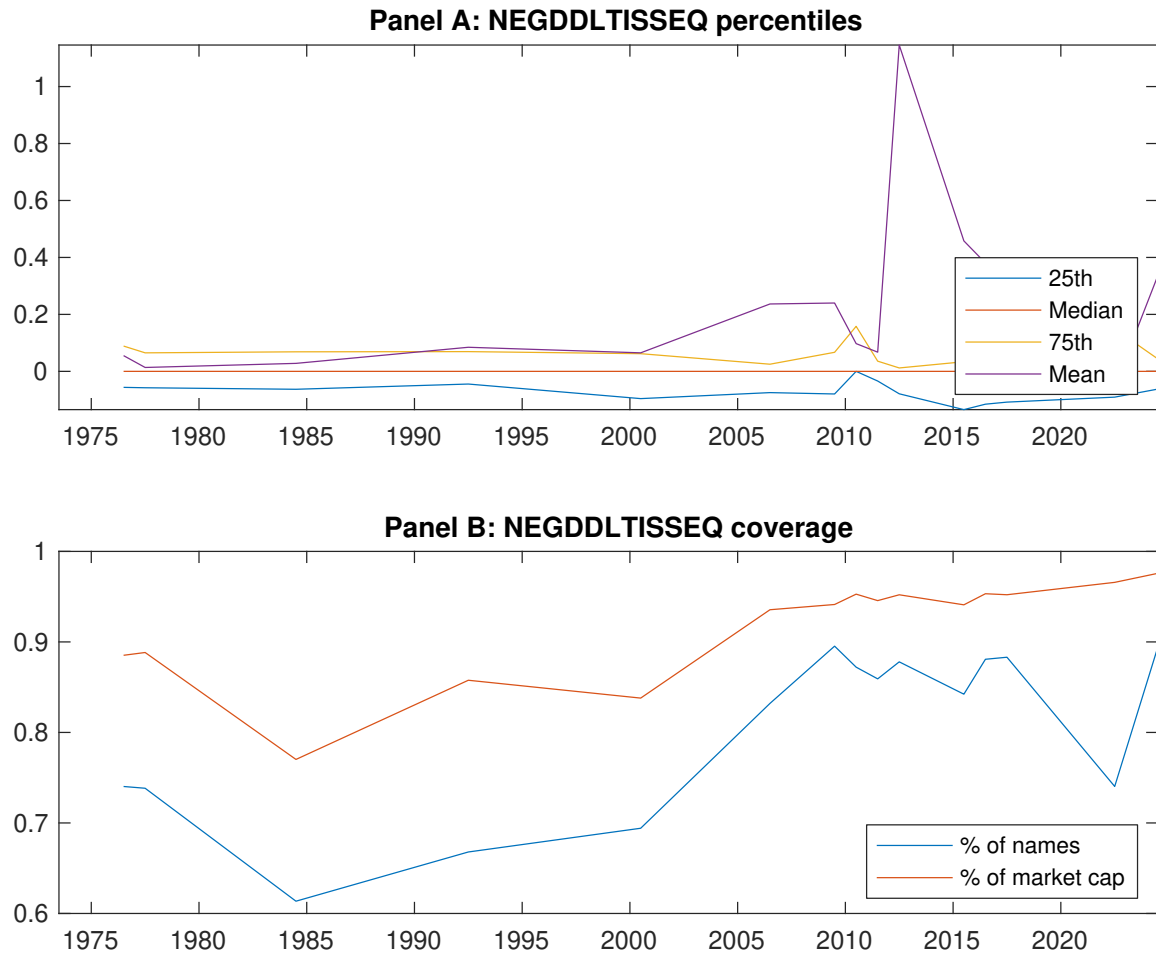


Figure 1: Times series of EDP percentiles and coverage.
This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.57]	0.66 [3.66]	0.64 [3.24]	0.73 [4.10]	0.82 [4.15]	0.27 [3.81]
α_{CAPM}	-0.18 [-3.14]	0.05 [0.98]	-0.02 [-0.41]	0.12 [2.67]	0.15 [2.82]	0.33 [4.68]
α_{FF3}	-0.20 [-3.57]	0.00 [0.06]	0.04 [0.81]	0.11 [2.37]	0.13 [2.45]	0.33 [4.66]
α_{FF4}	-0.17 [-2.97]	0.02 [0.42]	0.07 [1.33]	0.08 [1.74]	0.11 [2.14]	0.28 [3.97]
α_{FF5}	-0.20 [-3.49]	-0.05 [-1.06]	0.10 [1.87]	0.03 [0.70]	0.05 [0.92]	0.24 [3.48]
α_{FF6}	-0.17 [-3.08]	-0.03 [-0.71]	0.12 [2.19]	0.02 [0.36]	0.04 [0.82]	0.22 [3.07]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	1.09 [83.34]	0.98 [95.31]	0.97 [75.45]	0.97 [90.95]	1.04 [86.06]	-0.05 [-2.98]
β_{SMB}	0.11 [5.68]	-0.11 [-7.02]	-0.03 [-1.35]	-0.03 [-2.12]	0.13 [7.21]	0.02 [0.79]
β_{HML}	0.08 [3.04]	0.14 [7.11]	-0.14 [-5.80]	-0.02 [-0.74]	-0.05 [-2.05]	-0.12 [-3.95]
β_{RMW}	0.10 [3.75]	0.12 [5.94]	-0.08 [-3.01]	0.08 [4.06]	0.12 [4.98]	0.02 [0.69]
β_{CMA}	-0.14 [-3.60]	0.03 [1.01]	-0.11 [-3.06]	0.18 [5.74]	0.17 [4.68]	0.30 [6.35]
β_{UMD}	-0.04 [-2.76]	-0.02 [-2.40]	-0.03 [-2.28]	0.02 [2.33]	0.01 [0.58]	0.04 [2.64]
Panel C: Average number of firms (n) and market capitalization (me)						
n	649	551	1069	605	621	
me (\$10 ⁶)	1495	3003	2305	3111	1439	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.81]	0.33 [4.68]	0.33 [4.66]	0.28 [3.97]	0.24 [3.48]	0.22 [3.07]
Quintile	NYSE	EW	0.23 [4.86]	0.25 [5.44]	0.24 [5.19]	0.23 [4.74]	0.21 [4.54]	0.20 [4.29]
Quintile	Name	VW	0.26 [3.68]	0.32 [4.74]	0.33 [4.78]	0.27 [3.98]	0.27 [3.88]	0.23 [3.37]
Quintile	Cap	VW	0.24 [3.65]	0.29 [4.37]	0.29 [4.39]	0.23 [3.47]	0.20 [3.11]	0.16 [2.52]
Decile	NYSE	VW	0.31 [3.23]	0.38 [3.99]	0.38 [4.01]	0.33 [3.42]	0.30 [3.09]	0.27 [2.75]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.99]	0.26 [3.70]	0.26 [3.71]	0.23 [3.35]	0.19 [2.74]	0.17 [2.47]
Quintile	NYSE	EW	-0.03 [-0.48]					
Quintile	Name	VW	0.20 [2.83]	0.26 [3.80]	0.26 [3.80]	0.23 [3.38]	0.21 [3.08]	0.18 [2.76]
Quintile	Cap	VW	0.19 [2.89]	0.24 [3.64]	0.24 [3.64]	0.21 [3.17]	0.17 [2.60]	0.14 [2.23]
Decile	NYSE	VW	0.24 [2.50]	0.30 [3.18]	0.31 [3.19]	0.28 [2.89]	0.24 [2.46]	0.21 [2.22]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDP Quintiles					EDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.39]	0.87 [3.20]	0.97 [3.58]	0.85 [3.06]	0.80 [2.87]	0.14 [1.49]	0.17 [1.91]	0.16 [1.78]	0.11 [1.14]	0.09 [0.99]	0.06 [0.61]
	(2)	0.71 [2.66]	0.91 [3.61]	0.83 [3.32]	0.86 [3.52]	0.87 [3.43]	0.16 [2.14]	0.20 [2.57]	0.16 [2.21]	0.17 [2.29]	0.12 [1.64]	0.14 [1.77]
	(3)	0.79 [3.16]	0.78 [3.55]	0.85 [3.55]	0.80 [3.52]	0.90 [3.82]	0.11 [1.40]	0.16 [2.02]	0.16 [2.02]	0.12 [1.52]	0.15 [1.83]	0.12 [1.50]
	(4)	0.70 [3.03]	0.81 [3.95]	0.86 [3.88]	0.70 [3.39]	0.91 [4.19]	0.21 [2.88]	0.25 [3.43]	0.24 [3.23]	0.21 [2.75]	0.21 [2.81]	0.19 [2.47]
	(5)	0.45 [2.26]	0.62 [3.54]	0.59 [2.97]	0.65 [3.62]	0.79 [4.12]	0.33 [3.81]	0.37 [4.19]	0.39 [4.38]	0.32 [3.63]	0.31 [3.43]	0.26 [2.96]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDP Quintiles					EDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	391	394	393	393	389	37	34	32	33	35	
	(2)	107	107	107	108	107	61	62	60	61	61	
	(3)	76	77	76	77	76	108	109	105	106	108	
	(4)	64	65	64	65	64	230	238	230	234	231	
(5)	59	59	59	59	59	1384	2158	1902	2148	1585		

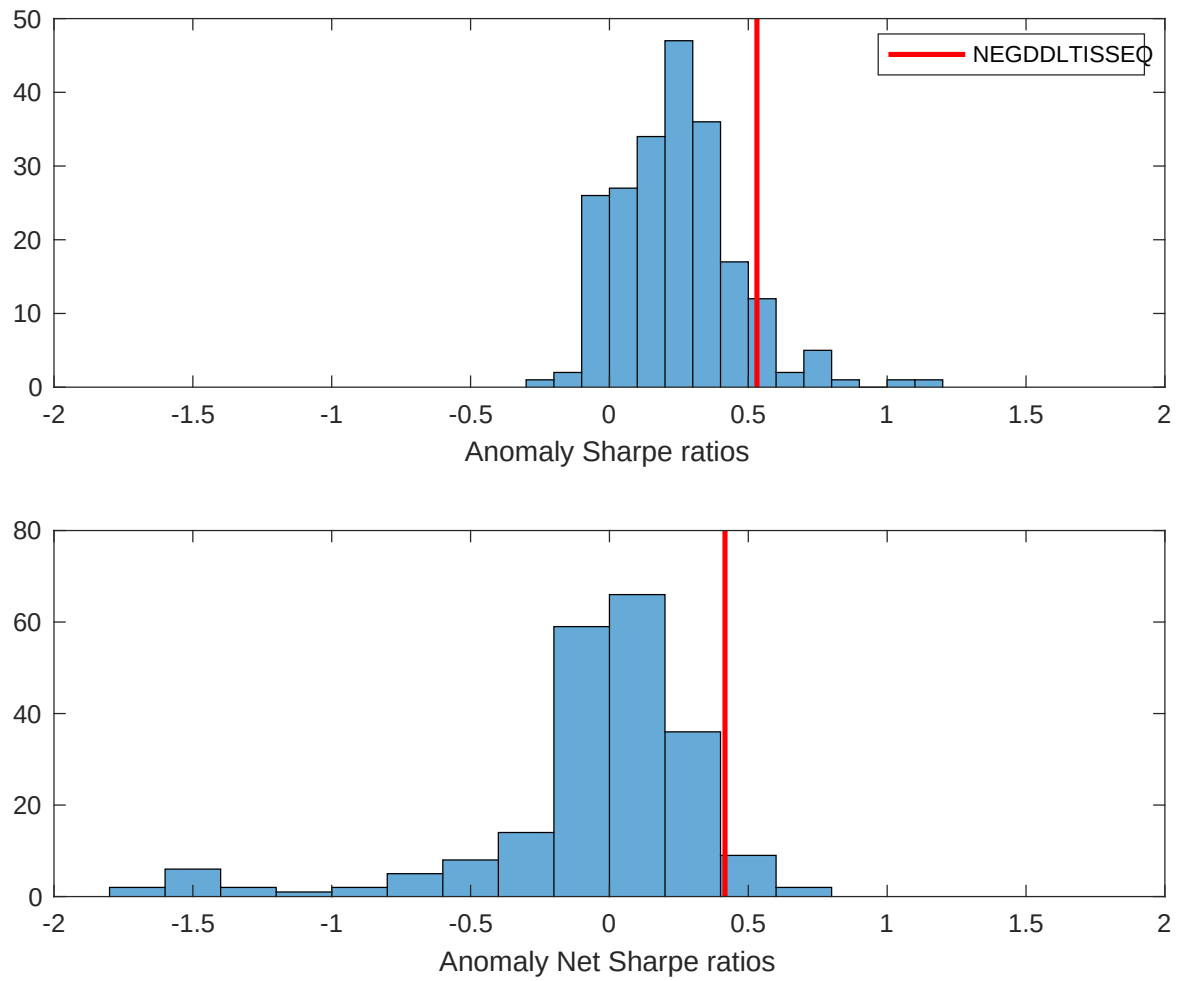


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

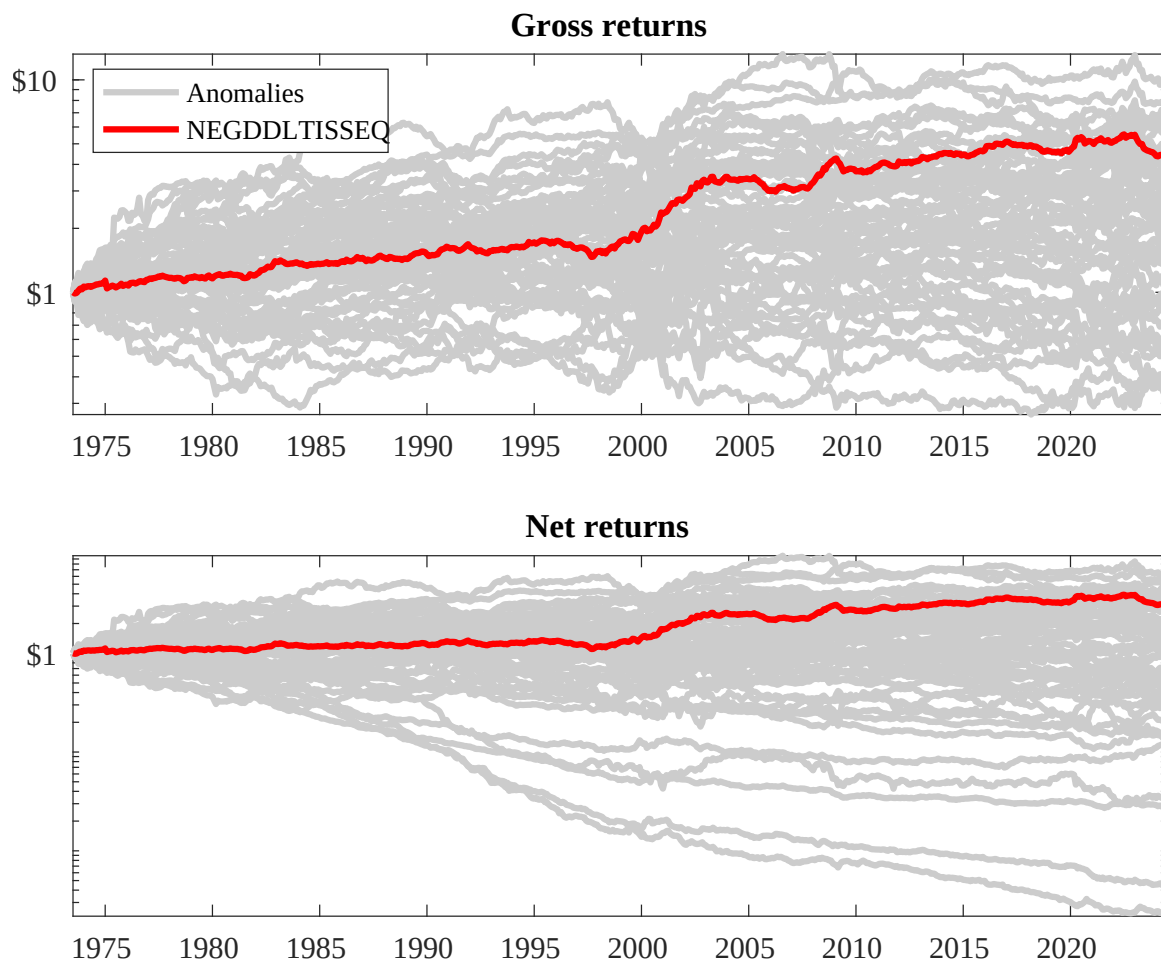
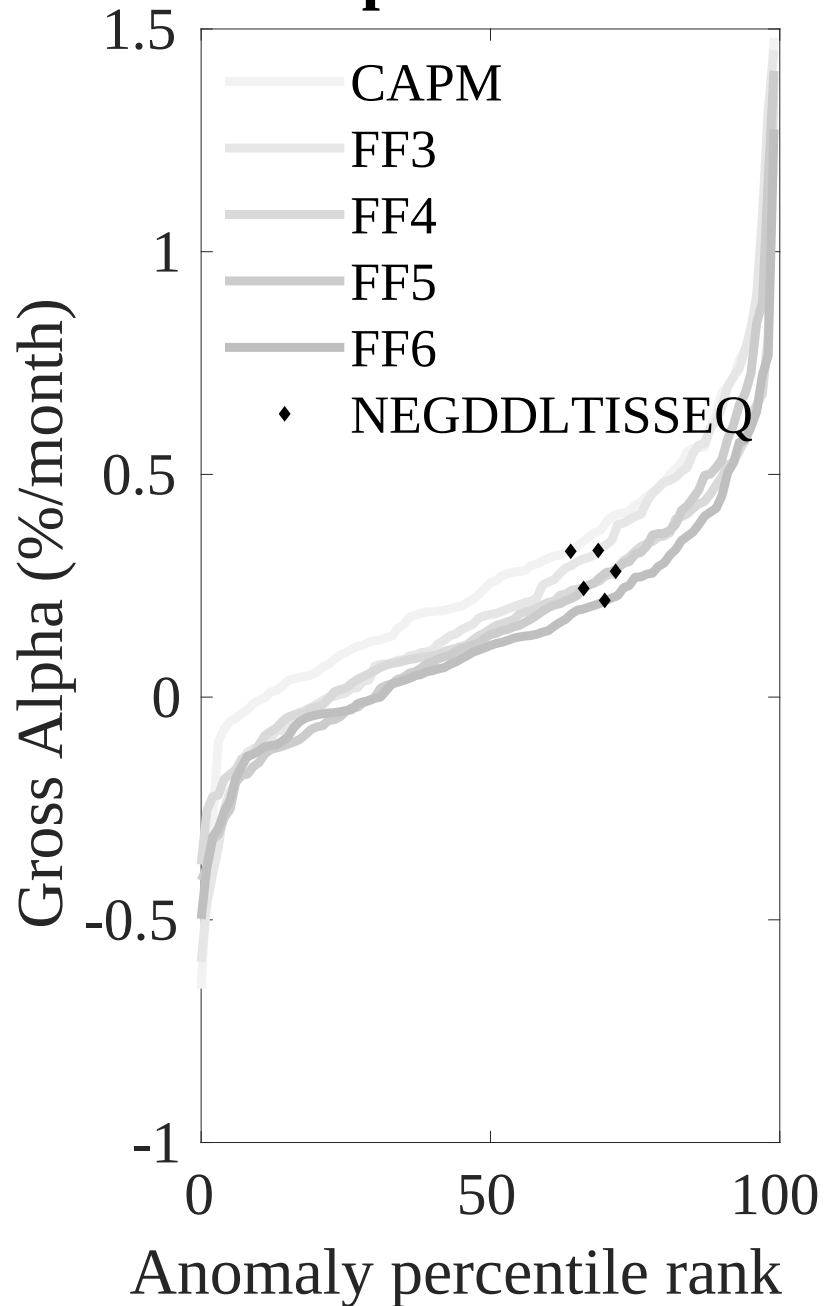


Figure 3: Dollar invested.

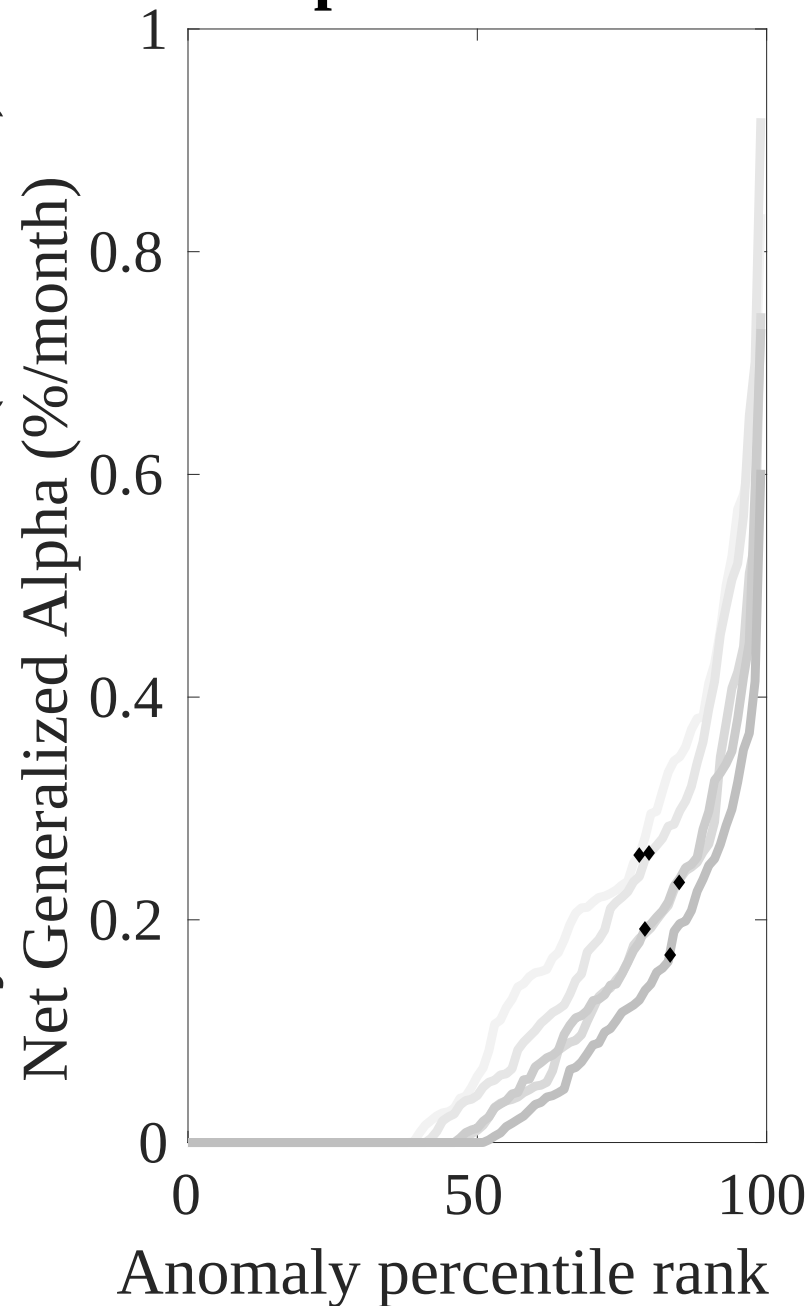
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



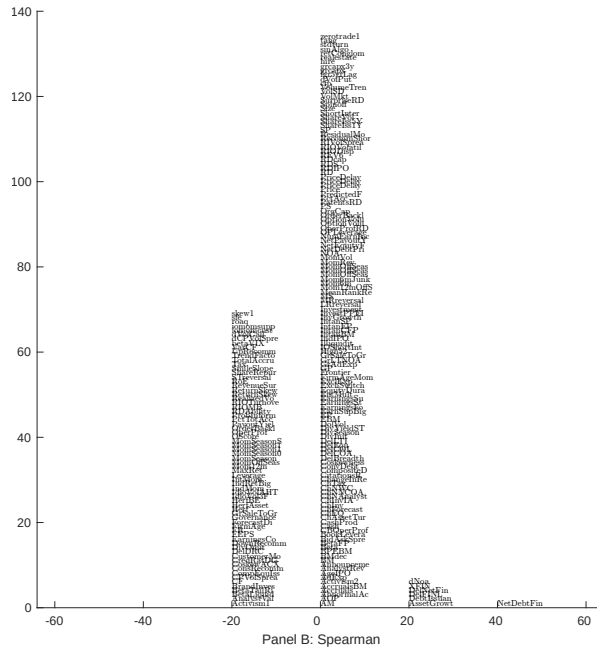
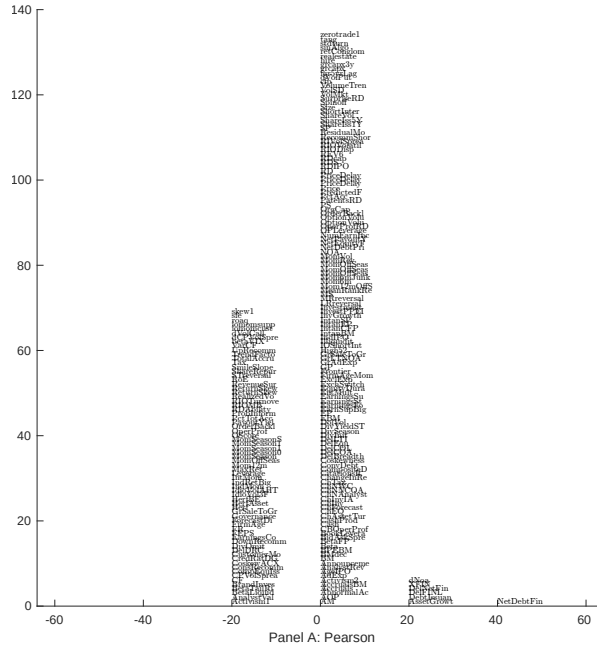


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

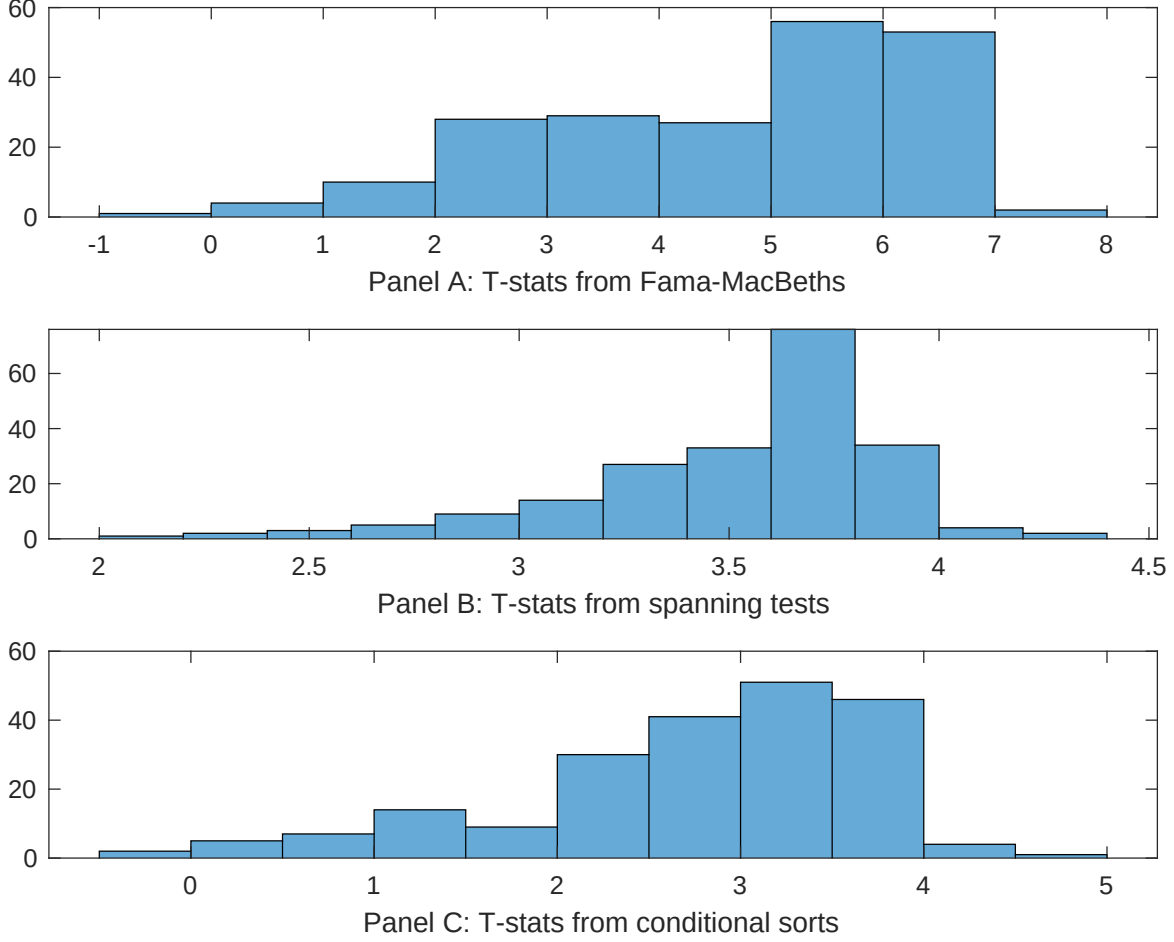


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.32]	0.13 [5.36]	0.14 [5.68]	0.14 [5.79]	0.13 [5.30]	0.14 [5.65]	0.15 [5.69]
EDP	0.56 [2.01]	0.45 [1.71]	0.78 [2.62]	0.36 [1.29]	0.12 [4.08]	0.30 [1.16]	0.10 [0.03]
Anomaly 1	0.20 [8.86]						0.98 [1.47]
Anomaly 2		0.17 [9.40]					-0.11 [-2.24]
Anomaly 3			0.19 [6.33]				0.10 [1.73]
Anomaly 4				0.10 [8.63]			0.40 [1.76]
Anomaly 5					0.42 [7.10]		0.45 [0.81]
Anomaly 6						0.14 [9.84]	0.94 [5.23]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.18 [2.73]	0.18 [2.67]	0.19 [2.70]	0.20 [2.88]	0.21 [2.99]	0.19 [2.79]	0.19 [2.75]
Anomaly 1	20.49 [5.51]						13.34 [2.62]
Anomaly 2		17.00 [4.38]					8.55 [1.61]
Anomaly 3			12.49 [3.62]				7.78 [2.12]
Anomaly 4				7.65 [1.90]			3.60 [0.84]
Anomaly 5					7.63 [2.83]		7.13 [2.61]
Anomaly 6						4.28 [1.08]	-6.52 [-1.47]
mkt	-4.70 [-3.01]	-4.35 [-2.76]	-3.03 [-1.83]	-4.75 [-2.98]	-4.88 [-3.07]	-4.73 [-2.96]	-3.60 [-2.20]
smb	-1.23 [-0.51]	-1.16 [-0.48]	4.60 [1.74]	0.03 [0.01]	1.42 [0.59]	0.66 [0.27]	1.43 [0.52]
hml	-12.62 [-4.19]	-12.32 [-4.04]	-12.06 [-3.92]	-13.80 [-4.47]	-13.69 [-4.46]	-13.80 [-4.45]	-11.34 [-3.70]
rmw	-0.11 [-0.04]	0.61 [0.20]	-5.32 [-1.43]	2.17 [0.69]	3.25 [1.03]	2.21 [0.71]	-3.81 [-1.02]
cma	26.84 [5.74]	26.47 [5.52]	24.01 [4.66]	22.89 [3.40]	25.38 [4.85]	28.99 [5.26]	14.91 [2.17]
umd	3.08 [1.96]	2.93 [1.83]	4.32 [2.74]	4.57 [2.87]	3.62 [2.26]	4.27 [2.67]	2.30 [1.43]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	18	16	15	14	15	14	19

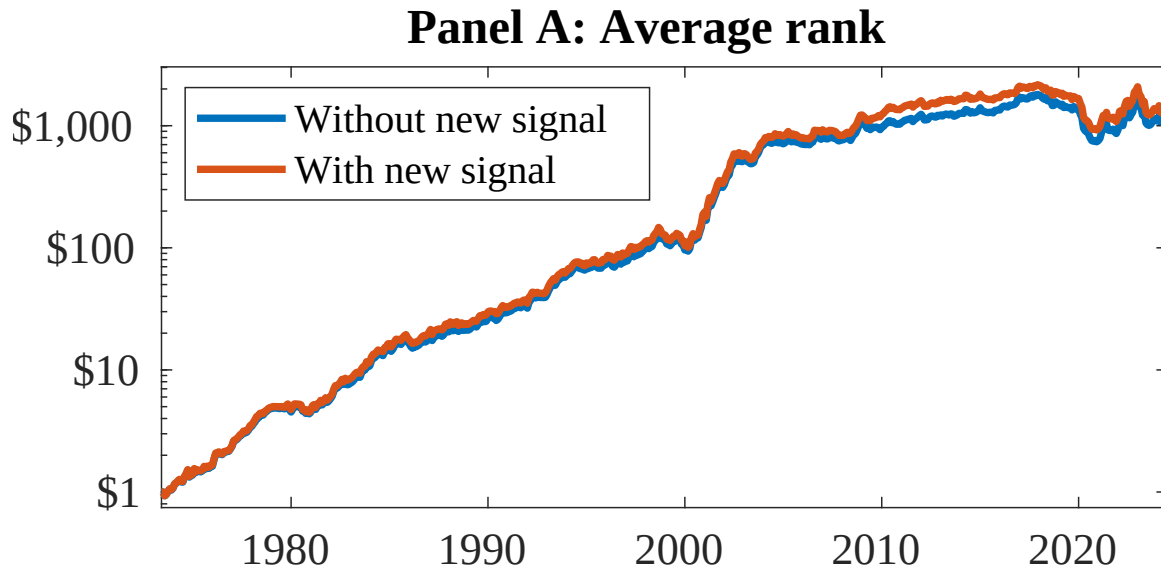


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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